

Tom Karingada

+1 224-202-0952 | tomkaringada04@gmail.com | [linkedin.com/in/tom-karingada](https://www.linkedin.com/in/tom-karingada) | github.com/tkaringa | tkaringa.github.io

Security Analyst and Penetration Tester specializing in IoT device security. Demonstrated expertise in leading end-to-end security assessments, executing complex attacks, and reverse engineering applications to uncover critical vulnerabilities. Highly proficient in translating rigorous compliance standards into actionable test plans, with a strong track record of guiding cross-functional teams and delivering high-value technical reporting.

EXPERIENCE

UL Solutions

Jun 2025 – Present

TS Security Analyst, Identity Management & Security / Device Security

Northbrook, IL

- Lead end-to-end security assessments across diverse domains, including IoT products, medical devices, and distributed energy resource platforms, evaluating hardware, mobile, and network architectures.
- Conducted rigorous wireless and protocol security testing across BLE, Wi-Fi, NFC, and USB interfaces, utilizing custom signal emulation, protocol fuzzing, and unencrypted traffic analysis to validate transport security.
- Authored Detailed Test Reports (DTRs) for stringent cybersecurity standards, translating requirements for EN 18031-1/2, UL 2900-1, and UL 2941 frameworks into evidenced functional assessments.
- Directed cross-functional remediation efforts by managing risk registers, and advising global engineering teams on secure architecture implementations and vulnerability resolution.

UL Solutions

May 2024 – Aug 2024

Software Engineering and Cybersecurity Intern

Northbrook, IL

- Shadowed cybersecurity assessments for connected products, documenting threats, and mitigations.
- Created training material for EN 18031, as well as mappings between EN 18031 and UL 2900.

Yaskawa America

May 2023 – Jul 2023

IT Intern

- Modified and deployed internal software systems, improving employee productivity by 8%.

TECHNICAL SKILLS

Security Testing: Vulnerability Analysis, IoT Testing, Mobile Security, Fuzzing, Reverse Engineering, Static Analysis

Tools & Platforms: Wireshark, Nmap, Metasploit, Ghidra, mtkclient, magiskboot, JADX/APKTool, MobSF, ESP32, AWS, Docker

Standards & Frameworks: EN 18031-1/2, UL 2900, and UL 2941

Programming: C/C++, Assembly, Python, JavaScript, Java, R

EDUCATION

University of Illinois Urbana-Champaign

Urbana-Champaign, IL

Master of Computer Science · GPA: 3.86

Aug 2025 – Present

University of Illinois Urbana-Champaign

Urbana-Champaign, IL

B.S. Computer Science & Astronomy · GPA: 3.73

August 2023 – May 2025

PROJECTS

[Morphologically-Informed Tokenization](#) | Python, NLP, BPE (Byte-Pair Encoding)

- A hybrid morphology and BPE tokenization pipeline for Malayalam, significantly improving model performance by reducing Out-Of-Vocabulary (OOV) rates by 71.7%.

[Thirayuka AI Information Retrieval and Text Classification System](#) | Python, NLP, BPE (Byte-Pair Encoding)

- Built an NLP search system using DistilBERT for a low-resource language, achieving 99% classification accuracy.

CERTIFICATIONS

CompTIA TestOut Security Pro — TestOut, December 2021

CompTIA TestOut Ethical Hacker Pro — TestOut, April 2022